DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

PROPOSED STUDY & EVALUATION SCHEME

FOR

III YEAR B.TECH.

(FOOD TECHNOLOGY)

ON

CHOICE BASED CREDIT SYSTEM (CBCS)

[EFFECTIVE FROM THE SESSION 2018-19]

B.TECH. (FOOD TECHNOLOGY)

3rd Year V-SEMESTER **Session- 2018-19**

SNo	Subject Code	Subject Name	L-T-P	Th/Lab Marks	Sessional		Total	Credit
					CT	TA		
1	RAS501	Managerial Economics	3---0---0	70	20	10	100	3
2	RAS502 /RUC501	Sociology / Cyber Security	3---0---0	70	20	10	100	3
3	RFT501	Fruits ,Vegetables and Plantation Products	3---0---0	70	20	10	100	3
4	RFT502	Principles of Food Preservation	3---**1**---0	70	20	10	100	**4**
5	RCH501	Chemical Reaction Engineering	3---0---0	70	20	10	100	**3**
6	RFT051 RFT053	DEPTT ELECTIVE COURSE-1	3---1---0	70	20	10	100	4
7	RFT551	Fruits Vegetables and Plantation Products Lab	0---0---2	50		50	100	1
8	RFT552	Food Preservation Lab	0---0---2	50		50	100	1
9	RFT553	Soft Computing Lab	0---0---2	50		50	100	1
10	RCH551	Chemical Reaction Engineering Lab	0---0---2	50		50	100	1
	TOTAL			**620**	**120**	**260**	**1000**	**24**

DEPARTMENT ELECTIVE COURSE-1:

1. **RFT051: Food Additives**
2. **RFT052:Frozen Foods and Cold Chain Management**
3. **RFT053:Technology of Beverages**

3rd Year VI- SEMESTER **Session- 2018-19**

SNo	Subject Code	Subject Name	L-T-P	Th/Lab Marks	Sessional		Total	Credit
					Test	Assig/ Att		
1	RAS601	Industrial Management	3---0---0	70	20	10	100	3
2	RUC601/ RAS602	Cyber Security/ Sociology	3---0---0	70	20	10	100	3
3	RFT601	Bakery Technology	3---0---0	70	20	10	100	3
4	RFT602	Dairy Technology	3---1---0	70	20	10	100	4
5	RCH604	Mass Transfer Operations	3---0---0	70	20	10	100	3
6	RFT061 RFT063	DEPTT ELECTIVE COURSE-2	3---1---0	70	20	10	100	4
7	RFT651	Bakery Technology Lab	0---0---2	50		50	100	1
8	RFT652	Dairy Technology Lab	0---0---2	50		50	100	1
9	RCH655	Mass Transfer Operations Lab	0---0---2	50		50	100	1
10	RFT654	Case Study*	0---0---2			100	100	1
	TOTAL			**570**	**120**	**310**	**1000**	**24**

DEPARTMENT ELECTIVE COURSE-2:

1. **RFT061:Cereals, Pulses and Oilseed Products**
2. **RFT062:Engineered, Fabricated and Textured Foods**
3. **RFT063:Food Process Engineering**

***Note: Case study pertaining to Food Industry, Processes, Safety audit etc**.

RFT-501 FRUITS, VEGETABLES AND PLANTATION PRODUCTS

L : T: P
3 : 0: 0

UNIT -I

Structural, Compositional and Nutritional aspects of fruits and vegetables. Physiological development: Growth, Maturation, Ripening and Senescence. Post harvest handling including controlled and modified storage. Techniques of processing and preservation of fruits and vegetables by refrigeration and freezing, canning and bottling, drying and dehydration.

UNIT -II

Technology of fruits and vegetable products: Juices and pulps, Concentrates and powders, Squashes and cordials. Beverage: Still and carbonated. James, Jellies and Marmalades. Preserves, candies and crystallized fruits. Tomato products: Puree, Paste, Ketchup, Sauce and soup. Chutneys, pickles and other products.

UNIT -III

Spices: Composition, Structure and characteristics. Preservation and processing of major and minor spices of India; whole spice, Spice powder, Paste and extracts, Spice oils and oleoresins. Composition, Structure ,characteristics & processing of cashew nut and other dry fruits

UNIT -IV

Composition, Production and processing of Tea leaves: Black tea, Green tea and Oolong tea. Instant tea. Production and processing of coffee cherries by wet and dry methods to obtain coffee beans, grinding, storage and preparation of brew, Soluble /Instant coffee, Use of chicory in coffee, decaffeinated coffee.

UNIT -V

Production, processing and chemical composition of cocoa beans. Cocoa Processes: Cleaning, roasting, alkalization, cracking and fanning, Nib grinding for cocoa liquor, cocoa butter and cocoa powder. Manufacturing process for chocolate: Ingredients, Mixing, Refining, Conching, Tempering, Moulding etc. to obtain chocolate slabs, chocolate bars. Enrobed and other confectionary products.

Text Books:

1. G. Lal, G.S. Siddappa and G.L. Tondan: Preservation of fruits & vegetables(2009), ICAR New Delhi
2. B. Shrilakshimi: Food Science (2003), New Age International.
3. Bernard. W. Minifie: Chocolate, Cocoa and Confectionary: Science and Technology(1989) Springer US
4. R.H.H. Wills et.al.: Postharvest : an introduction to the physiology and handling of fruit and vegetables (1981) Kensington, N.S.W. : New South Wales University Press
5. Norman N. Potter, Joseph H. Hotchkiss: Food science, 5th ed.New York : Chapman & Hall

RFT-502 PRINCIPLES OF FOOD PRESERVATION

L : T: P
3 : 1: 0

UNIT-I

Basic considerations: Aims and objectives of preservation & processing of foods, Characteristics of tissues and non-tissues foods, Degree of perishability of unmodified foods, Causes of quality deterioration and spoilage of perishable foods, intermediate moisture foods, wastage of foods.

UNIT -II

Preservation of foods by low temperatures:

(A) Chilling temperatures: Consideration relating to storage of foods at chilling temperatures, Applications and procedures, Controlled and Modified atmosphere storage of foods, Post storage Handling of foods.

(B) Freezing temperatures: Freezing process, Slow and fast freezing of foods and its consequence, other occurrences associated with freezing of foods. Technological aspects of pre freezing, Actual freezing, Frozen storage and thawing of foods.

UNIT -III

Preservation of foods by high temperatures: Basic concepts in thermal destruction of microorganisms D, Z, F values. Heat resistance and thermophilic microorganisms. Cooking, Blanching, Pasteurization and Sterilization of foods. Assessing adequacy of thermal processing of foods, General process of caning of foods, Spoilage in canned foods.

UNIT -IV

Preservation by water removal:

(a) Principles, Technological aspects and application of evaporative concentration process; Freeze concentration and membrane process for food concentrations.

(b) Principles, Technological aspects and application of drying and dehydration of foods, Cabinet, tunnel, belt, bin, drum, spray, vacuum, foam mat, fluidized-bed and freeze drying of foods.

UNIT -V

Principles, Technological aspects and application of sugar and salt, Antimicrobial agents, Biological agents, non ionizing and ionizing radiations in preservation of foods. Hurdle technology.

Text Books:

1. Marcus Karel, Owen R. **Fennema**, Daryl B. Lund: Physical principles of food preservation. New York, Basel Dekker, 1975.
2. Desrosier Norman W: The Technology of Food Preservation, AVI Pub. Co.
3. Norman N. Potter, Joseph H. Hotchkiss: Food science, 5th ed.New York : Chapman & Hall

RCH-501 CHEMICAL REACTION ENGINEERING

L : T: P
3 : 1: 0

UNIT 1

Rate of Reaction, Elementary and non-elementary homogeneous reactions, Molecularity and order of reaction, Mechanism of reaction, temperature dependency from thermodynamics, collision and activated complex theories. Integral and differential methods for analyzing kinetic data, interpretation of constant volume reactor, zero, first, second and third order reactions, half life period, irreversible reaction in parallel and series, catalytic reaction, auto catalytic reaction, reversible reactions.

UNIT 2

Interpretation of variable volume batch reactions for zero, first and second order reactions, design equation for batch, continuous stirred tank, plug flow reactors for isothermal reaction.

UNIT 3

Optimum reactor size, plug flow/mixed flow reactors in series and parallel, recycle reactor.

UNIT 4

Design of reactors for multiple reactions, parallel and series reactions. Temperature and pressure effects for single reaction.

UNIT 5

Residence time distribution of fluids in vessels, E, F and C curves, Dispersion model, Tank in series model. Non Isothermal PFR and CSTR, Safety issues in Non Isothermal Reactors.

Text Books:

1. Smith, J, M, "Chemical Engineering Kinetics", 3rd Edition, McGraw-Hill (1990).
2. Levenspiel, O., "Chemical Reaction Engineering", 3rd Edition, John Wiley (1998).
3. Fogler, H.S., 2016. Elements of chemical reaction engineering.

Reference Book:

1. Keith J. Laidler, "Chemical Kinetics" 3rd Edition, Pearson (2013)

DEPARTMENTAL ELECTIVE COURSE-1

RFT-051 FOOD ADDITIVES

L- T- P
3- 1- 0

UNIT I

Definitions of Food Additives, Classification and Functions, Legitimate uses of Additives in foods, Intentional & Non Intentional additives, Indirect food additives; Difference between Additives & Adulterants, Food uses and functions in formulations; Toxicological evaluation of food additives.

UNIT II

Uses & functions of:Acid, Base, Buffer systems, Salts and Chelating/Sequestering agents, Masticatory substances. Low calorie and non nutritive sweeteners, Polyols.

UNIT III

Antioxidants, Emulsifying and stabilizing agents, Anti-caking agents, Thickeners, Firming agents. Flour bleaching agents and Bread improvers.

UNIT IV

Anti microbial agents / Class I and Class II preservatives as per PFA Act. Clarifying agents. Gases and Propellants. Tracers and other additives.

UNIT V

Colours and Flavours (synthetic and natural) Types of flavours, Flavours generated during processing – reaction flavours, Stability of flavours during food processing, flavour emulsions; essential oils and oleoresins.

Text Books:

1. Branen AL, Davidson PM & Salminen S. 2001. Food Additives. 2nd Ed.Marcel Dekker.
2. Gerorge AB. 1996. Encyclopedia of Food and Color Additives. Vol. III.CRC Press.
3. Gerorge AB. 2004. Fenaroli's Handbook of Flavor Ingredients. 5th Ed.CRC Press.
4. Morton ID & Macleod AJ .1990. Food Flavours. Part A, BC. Elsevier.
5. Stephen AM. (Ed.). 2006. Food Polysaccharides and Their Applications.Marcel Dekker.
6. Fennema OR.1996. Food Chemistry. Marcel Dekker.
7. FSSAI , 2006.

RFT-052 FROZEN FOODS AND COLD CHAIN MANAGEMENT

L- T- P
3- 1- 0

UNIT -I

Fundamentals of freezing: Glass transition in frozen foods and biomaterial, microbiology of frozen foods, thermo-physical properties of frozen foods, Freezing load and freezing time calculations, Innovation in freezing process

UNIT -II

Facilities for the cold chain: freezing methods and equipment, cold store design and maintenance, transportation of frozen foods, retail display equipment and management, household refrigerators and freezers, monitoring and control of the cold chain.

UNIT -III

Quality and Safety of Frozen Foods: Quality and Safety of frozen meat and meat product, Quality and safety of frozen poultry and poultry products, Safety and quality fish, Shellfish and related products, Quality and safety of frozen vegetables, fruits, dairy products, ready meads , bakery products, Eggs and eggs products.

UNIT –IV

Monitoring and measuring techniques for quality and safety: Chemical measurements, sensory analysis of frozen foods, Food borne illnesses and detection of pathogenic microorganisms, self life prediction of frozen foods.

UNIT -V

Packaging of frozen foods: Introduction to frozen food packaging, plastic packaging of frozen foods, paper and card packaging of frozen foods, Packaging of frozen foods with other materials, Packaging machinery.

Text Books :

1. P.G.Smith, Introduction to Food Process Engineering, Springer.
2. O.R.Fennema : Principles of Food Science, AVI Publications
3. Mallett, C.P., Frozen Food Technology, Springer
4. Da-Wen Sun,Handbook of Frozen Food Processing and Packaging, Second Edition, CRC Press

RFT-053 TECHNOLOGY OF BEVERAGES

L- T- P
3- 1- 0

UNIT I.

Introduction, classification, Beverage industry in India, Traditional beverages; Effect of different ingredient on the quality of beverages; Preparation of syrups.

UNIT II

Manufacturing technology of mineral water and carbonated drinks; water quality, treatment and fortification process, Bottling, Packaging, storage and transportation.

UNIT III

Fruit beverages; Definition and preparation of squash, cordial, nectar, crush, alcoholic beverages; Milk beverages, Preparation of different beverages

UNIT IV

Selection and economics of different beverages packaging materials, selection, operation and maintenance of beverage machines / equipments, Automation in beverage industries.

UNIT V

Quality control and safety in beverage industries, Waste management in beverage industries, Sensory evaluation of beverages, Chemical and microbiological analysis of different beverages.

Text books:

1. L.F.Green: Developments in Soft Drinks Technology-I, Applied Science Publishers Ltd., London
2. Potter & Hotchkiss: Food Science, CBS Publishers & Distributors, New Delhi.
3. A.G.H. Lea, J.R.Piggot :Fermented Beverage Production, Springer-Science, Business Media, B.V

RFT-551 FRUITS, VEGETABLES AND PLANTATION PRODUCTS LAB

L : T: P
0 : 0 : 2

1. Preparation of Jam/Jelly and its preservation by sugar.
2. Preparation of tomato puree/ketchup and its preservation by chemical preservatives.
3. Preparation of Candied fruits.
4. Preparation of cordials and squash as per FPO specification.
5. Preparation of pickles.
6. Preparation of fermented Pickles (Sauerkraut).
7. Preparation of Fruit Preserves.
8. Preparation of Fruit Leather.
9. Preparation of Spice pastes.

Reference Books:

1. G. Lal, G.S. Siddappa and G.L. Tondan: Preservation of fruits & vegetables(2009), ICAR New Delhi
2. BIS Standards
3. FSSAI Manuals

RFT-552 FOOD PRESERVATION LAB

L : T: P
0 : 0 : 2

1. To check the adequacy of Blanching process
2. Extension of shelf life/ preservation of foods by use of low temperature.
3. Processing and preservation of Peas by use of high temperature.
4. To determine the effectiveness of sterilization process.
5. Preservation and processing of certain vegetables by drying and dehydration
6. Osmotic concentration/dehydration of certain fruits and vegetables
7. Preservation of fruit pulp with the help of Chemical preservatives.
8. Preservation of foods by Freezing (Frozen Foods).

Reference Books:

1. S.Ranganna: Handbook of Analysis and Quality Control for Fruit and Vegetable Products, Tata McGraw-Hill Education, 1986

RFT-553 SOFT COMPUTING LAB

L : T: P
0 : 0 : 2

Use of following Techniques in C/C++ Language

1. Solution of single non-linear algebraic equations by Newton Raphson method.
2. Solution of single non-linear equations by Regulafalsi method.
3. Solution of system of linear simultaneous by Gauss Elimination method.
4. Solution of system of linear simultaneous equation by gauss seidel method and successive over relaxation method.
5. Solution of single first order ordinary differential equations by fourth order Runge-Kutta method.
6. Solution of Heat equations (Parabolic equations) by finite difference method.
7. Solution of Laplace equations (elliptic equation) by finite difference method.
8. Solution of wave equations (Hyperbolic equation) by finite difference method.
9. Finding Newton's interpolatory polynomial for n points.
10. Finding Newton's interpolatory polynomial based on finite difference table for n points.
11. Simpson's 3/8-rule.

RCH-551 CHEMICAL REACTION ENGINEERING LAB

L : T: P
0 : 0 : 2

1. Find out kinetic constant and study conversion of a given reaction in a batch reactor
2. Find out kinetic constant and study conversion of a given reaction in a plug flow reactor
3. Find out kinetic constant and study conversion of a given reaction in a CSTR
4. Study and operation of an adiabatic batch reactor
5. Study of a reversible reaction in a batch reactor
6. To determine energy of activation of reaction of ethyl acetate with sodium hydroxide
7. Find out specific rate contant and activation energy of a reaction in a plug flow reactor
8. To determine reaction equilibrium constant of reaction of acetic acid with ethanol.

9. To determine changes in free energy, enthalpy and entropy for the reaction of potassium iodide with iodine.
10. Study and operation of a cascade CSTR

RFT-601 BAKERY TECHNOLOGY

L : T: P
3 : 0: 0

UNIT-I

Wheat flour and wheat flour treatments – Grade of flour, constituents of flour – ageing of flour – Tests for flour quality. Yeast : Characteristics, Preparation, Handling & Storage, Adequacy for use in bakery industry. Ingredients, Technology and quality parameters for baked products: Bread, Biscuits and cakes.

UNIT-II

Bakery equipment and machinery. Different types of Mixers, kneaders and cutters. Different types of ovens. Packaging machinery for bread and biscuits. Quality control in bakery industry. Quality control of raw materials. Quality control of finished products. Quality control of packaging materials.

UNIT-III

Technology of bread making Different methods. Process steps and their significance. Characteristics of good bread. Defects in bread their causes and remedies.

UNIT-IV

Technology of Cakes Manufacture. Different cake making processes. Sugar batter method, Flour batter method ,Modified sugar batter method Whipping method, Blending method etc. Process steps and their significance. Importance of baking time and temperature. Recipe balancing .Defects in cakes, their causes and remedies.

UNIT-V

Biscuits. Definition and types. Fermented dough biscuits. Cookies. Types of cookies and their manufacture. Cream biscuits. Process steps and their significance. Defects in biscuits their causes and remedires.

Text Books :

1. E.J Pyler: Baking Science and Technology: Vol.1 & 2, 3rd Edition, Sosland, 1988
2. Samuel A.Matz: Bakery Technology and Engineering, Springer US
3. Samuel A.Matz: Cookie and Cracker technology, AVI Publications
4. H. Faridi:The Science of Cookie and Cracker Production, CBS Publishers & Distributors, N Delhi

RFT-602 DAIRY TECHNOLOGY

L : T: P
3 : 1: 0

UNIT-I

Fluid Milk: Composition of milk and factor affecting it. Physico-chemical characteristics of milk and milk constituents. Production and collection , cooling and transportation of milk. Packaging storage and distribution of pasteurized milk.

UNIT -II

Whole, Standardized, Toned, Double toned and skim milk. Test for milk quality and Adulteration. UHT processed milk, flavored, Sterilized milk. Cleaning and sanitization of dairy equipments. Definition, Classification, Composition and physico-chemical properties of cream. Production processes and quality control.

UNIT -III

Butter: Definition, Classification, Composition and methods of manufacture, Packaging and storage. Butter oil/Ghee. Ice cream: Definition, Classification and Composition, Constituents and their role. Preparation of mixes and freezing of Ice cream, Overrun, Judging, Grading, and defects of Ice cream.

UNIT -IV

Evaporated and Condensed milk: Method of manufacture, Packaging and storage. Defects, Causes, and prevention. Roller and Spray Drying of milk solids. Instantization. Flow ability, Dustiness, Reconstituability, Dispersability, Wet ability, Sink ability and appearance of milk powders.

UNIT -V

Byproducts of Dairy Industry and their effective utilization. Manufacture of casein, Whey protein, Lactose from milk and their use in formulated foods. Quality Control tests in Dairy industry.

Text Books :

1. Sukumar De: Outlines of Dairy Technology, Oxford University Press, 1980
2. .Aneja, Mathur, Chandan & A.K.Bannerji: Technology of Indian Milk Products: Dairy India Publication
3. Shivashraya Singh: Dairy Technology: Vol.01: Milk and Milk Processing, New India Publishing Agency, 2013
4. Meghwal. Goyal and Chavan: Dairy Engineering: Advanced Technologies and their applications, apple Academic Press, Canada.

RCH-604 MASS TRANSFER OPERATIONS

L : T: P
3 : 0: 0

UNIT -I

Basic Principles of mass transfer: Molecular diffusion in fluids, mass transfer coeeficients, Interphase mass transfer. Vapour pressure, enthalpy, absolute humidity, dew point, etc., Unsaturated vapour gas mixtures.

UNIT -II

Gas absorption: Countercurrent, co-current, multistage continuous contact operations.Distillation : Entrainment, pressure drops , flooding , transfer coefficients and relative volatility. Macabe Thiele and Ponchon method for binary component distillation of azeotropes. Flash vaporization and Multicomponent distillation.

UNIT -III

Diffusion: Molecular and turbulent diffusion , diffusion coefficient. Fick's law of diffusion, measurement and estimation of diffusivity. Diffusion in multicomponent gas mixtures. Diffusion in solids: Molecular, Knud sen and surface diffusion. Inter phase mass transfer: Mass transfer coefficients, diffusion between phases, equilibrium solubility of gases in liquids. Mass transfer theories. Mass transfer in fluidized beds

UNIT -IV

Adsorption and Stripping,Equipments, gas-liquid equilibria, Henry's law, selection of solvent, absorption in tray column, graphical and analytical methods. Adsorption in packed columns. HTU, NTU & HETP concepts, design equations for packed column.

UNIT -V

Drying: Batch and freeze drying, rotary driers. Surface vs diffusion controlled operations. Solid-gas equilibria, definitions of moisture contents, types of batch and continuous dryers, rate of batch drying , time of drying, mechanism of batch drying, continuous drying.

Text Books:

1. Treybal, R "Mass Transfer Operations", 3rd ed. New York: McGraw-Hill, (1980).
2. Sherwood T. K., Pigford R. L. and ilke P. "Mass Transfer" McGraw Hill (1975).
3. Foust A. S. et.al., "Principles of Unit Operations" John Wiley (1980).
4. Geankoplis, C.J.. "Transport Processes and Unit Operations", 3rd ed. Prentice Hall. (1993)

DEPARTMENTAL ELECTIVE COURSE-2

RFT-061 CEREALS, PULSES AND OILSEED PRODUCTS L- T- P 3- 1- 0

UNIT-I

Composition, Structure and Processing characteristic of Cereal grains, Legumes and oilseeds, Post harvest, Post processing practices for their safe storage. Parboiling and Milling of paddy, Quality characteristics, Curing and aging of rice, Processed rice products.

UNIT-II

Wheat and its quality characteristics for milling into flour and semolina, Flour milling, Turbo grinding and air classification, Flour grades and their suitability for baking purposes, Assessment of flour quality and characteristics, Milling of Durum wheat, Macaroni products.

UNIT-III

Dry and Wet milling of corn, Starches and its conversion products, Cornflakes Manufacture. Malting of barley

UNIT-IV

Milling of legume-pulses by traditional and improved processes. Pearling of Millets.

UNIT-V

Processing of oil seeds for direct use and consumption, Oil and protein products. Processing of extracted oil. Refining, Hydrogenation, Interestrification. Processing of deoiled cake into protein concentrates and isolates, Textured protein, Functional protein preparations. Peanut butter, Margarine and Spread.

Text Books:

1. C.F.T.R.I. Mysore Manuals on Rice and its Processing
2. S.A.Matz : Cereal Technology, PAN-TECH International Incorporated, 1999
3. Y.Pomeranz: Cereals and Cereal Products, Wiley - VCH Verlag GmbH & Co. KGaA
4. S. A. Matz. :Cereal Technology. , Ed. Avi Publishing Co., Westport
5. S. A. Matz: Chemistry and Technology of Cereals as Food and Feed, AVI Publications
6. Chakravarti A. 1988. Postharvest Technology of Cereals, Pulses and Oilseeds. Oxford and IBH, New Delhi.

RFT-062 ENGINEERED, FABRICATED AND TEXTURED FOODS L- T- P 3- 1- 0

UNIT- I

Extruders. Single Screw & Multiple Screw Extruders. Extrusion process. Extrusion cooking. Physical & Chemical Changes during Extrusion Process.

UNIT -II

Textured vegetable protein products. Puffing Gun ,Puffed Products. Meat Analogues., Imitation Paneer

UNIT -III

Fabricated RTS Beverages, Bakery Products, Margarine, Peanut Butter, Imitation Milks Designer Lipids etc.

UNIT- IV

Weaning Foods/ Baby Foods. Therapeutic Foods . Geriatric Foods.

UNIT- V

Technology and manufacture of Macaroni, Pasta, Noodles, Vermicelli etc

Text Books :

1. Zeki Berk, Food Process Engineering and Technology, Academic Press
2. Shubhangini A. Joshi Nutrition and Dietetics, Tata McGraw-Hill Education, 2010
3. S.A.Matz, Cereal Technology, CBS Publishers

RFT-063 FOOD PROCESS ENGINEERING

L- T- P
3- 1- 0

UNIT -I

Introduction , Concept of Unit operation, Units and dimensions, Unit conversions, dimensional analysis, Mass and Energy Balances in various food processing operations. Related numericals
Grinding and mixing Principle and equipments used in food industry

UNIT -II

Fluid Flow in food Processing , Liquid Transport systems ,Properties of Liquids , Newton's Law of Viscosity , Principle of Capillary tube and rotational viscometer ,Properties of Non-Newtonian fluids, Flow characteristics, Reynolds Number, Bernoulli"s Equation , Concept of Flow Measurement devices ,Related basic numericals

UNIT –III

Calculations pertaining to Heat Transfer and Flow properties of liquid foods in Transportation and Processing. Kinetics of Chemical and Biochemical Reactions in Foods. Thermal stabilization of foods.

UNIT –IV

Refrigeration and Freezing ,Concept and selection of a refrigerant , Freezing time calculation, Frozen food storage , Refrigeration requirements during storage, Related basic numericals

UNIT –V

Important considerations for designing of food plants. Types of layout.

Text Books :

1.Albert Ibarz, Gustavo V. Barbosa-Canovas,Introduction to Food Process Engineering, CRC Press.
2. Smith P.G., Introduction to Food Process Engineering
3. Romeo T.Toledo, Fundamentals of Food Process Engineering, 3rd Edition, Springer.

RFT-651 BAKERY TECHNOLOGY LAB

L : T: P
0 : 0 : 2

1. Preparation of Bread/ Test Baking.
2. Preparation of Sweet Buns
3. Preparation of Pizza base.
4. Preparation of Biscuits
5. Preparation of Nan-Khatai
6. Preparation of Cookies
7. Preparation of Cakes (Eggs/ Eggless)
8. Preparation of Pastries
9. Preparation of Laminated and Puffed products

Reference Books:

1. S.C Dubey: Basic Baking: Science and Craft, Publisher, S C Dubey, 1980
2. Betty Crocker: Baking Basics, Wiley, 2009

RFT-652 DAIRY TECHNOLOGY LAB

L : T: P
0 : 0 : 2

1. Plateform Tests of milk [COB,MBR Test, Alcohol Test, Sediment Test]
2. Determination of Fat content in Milk and Milk Products
3. Determination of SNF Content in Milk
4. Determination of Titratable Acidity in Milk
5. Determination of Overrun in Icecream
6. Quality Testing of Butter
7. Quality Testing of Butter oil / Ghee
8. Analysis of Adulteration in Milk and Milk products

Reference Books:

1. Ramesh C. Chandan: Dairy-based Ingredients, Eagan Press, 1997
2. Sukumar De: Outlines of Dairy Technology, Oxford University Press, 1980
3. .Aneja, Mathur, Chandan & A.K.Bannerji: Technology of Indian Milk Products: Dairy India Publication

RCH-655 MASS TRANSFER OPERATIONS LAB

L : T: P
0 : 0 : 2

1. Separation factors of the experiments with differential distillation.
2. Separation factors of the experiments with flash vaporization.
3. Separation factors of the experiments with vapour liquid equilibrium.
4. Separation factors of the experiments with liquid – liquid extraction.
5. Separation factors of the experiments with solid –liquid extraction.
6. Separation factors of the experiments with ion exchange.
7. Separation factors of the experiments with membrane separation.
8. Studies on Bubble cap/ tray/ fractional column.
9. Studies on Absorption/ Humidification/ Dehumidification columns.
10. Studies on crystallization and adsorption.

RFT-654 CASE STUDY

L : T: P
0 : 0 : 2

Case study pertaining to Food Industry, Processes, Safety audit etc. shall be given to the students to carry out extensive search of literature/surveys and prepare a report for assessment followed with Seminar on the same.